

CSCE 645 GEOMETRIC MODELING · MODULE 10 · LECTURE NOTES

Simplification and Remeshing

Fewer triangles, better triangles.

MODULE OUTCOMES

- Implement quadric error metric simplification.
- Preserve attributes and seams through simplification.
- Explain why mesh quality affects operator behaviour.
- Implement isotropic remeshing by local operations.
- Explain Delaunay and anisotropic remeshing.

1 Simplification

Reduce triangle count while keeping the shape. The standard approach is greedy: repeatedly perform the cheapest edge collapse until the target is met.

- 1 Compute a **cost** for every edge and insert into a priority queue.
- 2 Pop the cheapest. Check the **link condition** and the **normal-flip test** (Module 02); skip if either fails.
- 3 Collapse, placing the merged vertex at the position the metric says is optimal.
- 4 Recompute costs for the affected edges and update the queue.
- 5 Repeat.

With a heap this is $O(n \log n)$. All of the design is in the cost function, and the standard answer has been the same since 1997.

1.1 Quadric error metrics

Garland and Heckbert's insight: associate with each vertex a **quadric** — a 4×4 symmetric matrix Q encoding the sum of squared distances to the planes of the faces originally incident to it.

$$\text{error}(\mathbf{v}) = \mathbf{v}^T \mathbf{Q} \mathbf{v} \quad (\mathbf{v} \text{ in homogeneous coordinates})$$

QUADRICS ADD, AND THAT IS THE WHOLE TRICK

When two vertices merge, the quadric of the result is simply $Q_a + Q_b$. No list of original planes needs to be retained; the 4×4 matrix carries all the information forward.

So the cost of any candidate collapse is computable in constant time, and the optimal position for the merged vertex is the solution of $\nabla(v^T Q v) = 0$ — a 4×4 linear solve. If the system is singular (a flat or symmetric configuration), fall back to the edge midpoint.

Without additivity, every collapse would require re-examining the original geometry, and the algorithm would be quadratic.

WHY THIS PRESERVES SILHOUETTES WITHOUT BEING TOLD TO

In a flat region every incident plane is the same plane, so a merged vertex anywhere on it has error near zero. These edges collapse first — correctly, because flat regions need few triangles.

At a sharp edge or a high-curvature ridge, the incident planes disagree strongly. Any merged position is far from at least one of them, so the cost is high and the edge survives.

The metric therefore concentrates triangles exactly where the shape requires them, with no feature detection and no parameters. That is why it has been the standard for nearly thirty years.

2 Attributes: where naive simplification fails

A mesh in a real pipeline carries more than positions: texture coordinates, normals, vertex colours, skinning weights, material assignments, lightmap coordinates.

COLLAPSING ACROSS A UV SEAM DESTROYS THE ATLAS

Two vertices at the same 3D position on either side of a UV seam have *different* texture coordinates. They are the same point on the surface and different points in the parameterisation.

A geometry-only simplifier sees one position and merges them, which silently destroys the seam and sends large regions of the texture to the wrong place. The geometry still looks correct, which makes the bug memorable.

The same applies to normal splits at hard edges, to material boundaries, and to skinning weight discontinuities.

Strategy	How	Trade-off
Forbid	Mark seam and boundary edges; never collapse them.	Safe and simple. Seams become over-tessellated relative to their surroundings at aggressive reduction ratios.

Strategy	How	Trade-off
Extend the quadric	Enlarge Q to include attribute dimensions, so error measures geometric <i>and</i> attribute deviation.	Principled and the best quality. Larger matrices, more arithmetic, more code.
Weight heavily	Allow seam collapses but multiply their cost by a large factor.	A middle path; a single tunable parameter that artists can override per asset.

2.1 Level of detail in practice

Approach	Mechanism	Assessment
Discrete LOD	Prebuild 3–5 levels; switch by screen-space size.	Simple and predictable. Pops visibly at the switch; needs dithered or cross-faded transitions.
Progressive mesh	Store the ordered sequence of collapses; reverse them to refine.	Continuous and smooth. Complex runtime and awkward on the GPU.
Normal map baking	Simplify the geometry, then bake the high-resolution surface's normals into a texture on the low-resolution mesh.	The standard pipeline. Recovers almost all shading detail. The silhouette is still simplified, which is the limitation CSCE 641 Module 08 identified.
Virtualised geometry	Cluster-level LOD selected per frame.	Eliminates both popping and manual LOD authoring, at the cost of a substantial runtime system.

3 Remeshing

Simplification reduces triangle count. **Remeshing** improves triangle *quality*, usually at a similar count, and it matters far more than it appears to.

QUALITY DETERMINES WHETHER YOUR OPERATORS WORK

Module 07: cotangent weights are negative for obtuse angles, so the discrete Laplacian loses the maximum principle and smoothing can move vertices the wrong way.

Module 09: negative weights violate Tutte's hypothesis, so the bijectivity guarantee fails and parameterisations develop flipped triangles.

Module 08: the explicit smoothing stability limit scales with the *shortest* edge, so a single sliver forces a tiny time step for the whole mesh.

Numerics generally: badly shaped triangles produce badly conditioned matrices, and iterative solvers converge slowly or not at all on them.

Remeshing is therefore not tidying up. It is making the rest of the course's machinery function.

3.1 Isotropic remeshing

Given a target edge length L , repeat 5-10 times:

1. SPLIT every edge longer than $(4/3)L$
2. COLLAPSE every edge shorter than $(4/5)L$
3. FLIP edges where doing so reduces total valence deviation from 6
4. SMOOTH tangentially (Module 08), then project each vertex back onto the original surface

Every ingredient is from earlier modules: the three local operations of Module 02, the tangential smoothing of Module 08. Only the schedule and the projection step are new.

- **The 4/3 and 4/5 thresholds** are chosen so that a newly split edge is not immediately a collapse candidate and vice versa. Choosing them naively — say, $1.5L$ and $0.5L$ with the wrong ratio — produces an algorithm that oscillates and never converges.
- **Projection back onto the original surface** is essential. Without it, repeated smoothing shrinks and distorts the shape (Module 08). With it, vertices redistribute across the surface while the surface itself is preserved.

3.2 Why valence 6

Module 01 established that Euler's formula forces the *average* vertex valence of a closed triangle mesh to be approximately 6. Valence 6 everywhere is therefore the most uniform configuration the topology permits, and a mesh close to it has triangles close to equilateral — which is exactly the condition that keeps cotangent weights positive and matrices well conditioned.

An edge flip is accepted when it reduces the total deviation from 6 across the four affected vertices. Topology sets the target; the algorithm merely pursues it.

4 Other remeshing goals

Goal	Method	Use for
Isotropic	Uniform target edge length; valence 6; tangential smoothing.	Simulation meshes, finite elements, and anything where operator quality matters.
Adaptive	Target edge length proportional to local feature size or inverse curvature.	Detail concentrated where the surface is curved, coarse where it is flat.
Anisotropic	Triangles elongated along principal curvature directions (Module 07).	Approximating a surface to a given tolerance with the fewest triangles.
Quad remeshing	Compute a smooth cross field, then extract a quad layout aligned to it.	Subdivision control cages, animation-friendly topology. The hardest of these and the one artists most want.
Delaunay	Flip edges until the empty-circumcircle property holds.	Maximises the minimum angle, which is precisely the condition that avoids negative cotangent weights.

ANISOTROPIC REMESHING, AND THE TRADE IT MAKES

A cylinder is curved in one direction and perfectly flat in the other. Isotropic remeshing gives it equally sized triangles everywhere, which spends triangles along the flat direction where none are needed.

Anisotropic remeshing stretches triangles along low-curvature directions and compresses them across high-curvature ones, approximating the same surface to the same tolerance with substantially fewer triangles.

The price is that the triangles are deliberately far from equilateral, so every operator-quality benefit of §3 is given up. Use it when the mesh is for rendering and triangle budget is the constraint; use isotropic when the mesh is for computation.

INTRINSIC DELAUNAY TRIANGULATION

A more recent alternative worth knowing: rather than moving vertices, flip edges to obtain the **intrinsic** Delaunay triangulation — a different connectivity on the *same* vertex positions, where the edges are geodesics rather than straight lines in space.

This guarantees non-negative cotangent weights without altering the geometry at all, which means operators become well behaved on a mesh you are not allowed to change. Crane's DDG notes cover it, and it is increasingly the preferred answer when remeshing is not an option.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Garland & Heckbert — Surface Simplification Using Quadric Error Metrics (free)	The original paper. Short, clear, and still the basis of every production simplifier.
Botsch & Kobbelt — A Remeshing Approach to Multiresolution Modeling (free)	The isotropic remeshing algorithm of §3, including the threshold choice.
Polygon Mesh Processing — simplification and remeshing chapters	The comprehensive survey with implementation detail.
Sharp, Soliman & Crane — Navigating Intrinsic Triangulations (free)	Intrinsic Delaunay: fixing operator quality without moving a single vertex.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Implement quadric error simplification with the link condition and normal-flip checks. Reduce a 100,000-triangle scan to 10,000 and 1,000 and render all three at matched screen size.
- 2 Visualise which edges survive longest by colouring the mesh with collapse cost. Confirm that silhouettes and creases are preserved.
- 3 Simplify a textured mesh without seam handling and capture the destroyed UV atlas. Then add seam preservation and compare.
- 4 Measure simplification quality properly: compute the Hausdorff distance between the original and simplified meshes at several reduction ratios and plot it.
- 5 Implement isotropic remeshing. Plot the distribution of triangle angles before and after, and report the minimum angle in each.
- 6 Count obtuse triangles and negative cotangent weights before and after remeshing. Then re-run the Module 09 parameterisation on both and compare the number of flipped triangles.
- 7 Implement edge flipping to Delaunay and measure the improvement in minimum angle without moving any vertex.
- 8 Bake a normal map from a high-resolution mesh onto its simplified version. Render both at a distance where they should be indistinguishable, and find the distance at which the silhouette difference becomes visible.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Describe greedy edge-collapse simplification and the two validity checks it requires.
- 2 What does a quadric measure, and which property makes the algorithm efficient?
- 3 Why does quadric error preserve silhouettes without any feature detection?
- 4 What happens when a simplifier collapses across a UV seam, and name three remedies?
- 5 Give three specific ways poor triangle quality breaks algorithms from earlier modules.
- 6 List the four steps of isotropic remeshing and explain why the $4/3$ and $4/5$ thresholds are chosen as they are.
- 7 Why is valence 6 the target, and where does that number come from?
- 8 What does anisotropic remeshing gain and what does it give up?